

Title

- An Automated Data Deletion: A Secure Method for Multi-Cloud Security with Intrusion Detection System.

Problem Statement

- Cloud data can be exposed after unauthorized access.
- Organizations need automated response mechanisms.

Research Objectives

The objectives of our proposed research of automated data deletion system for multi-cloud environments extends across several key aspects. Following, are some of the key aspects of research significance:

- It will enhance the data security and mitigate the threats by automatically deleting sensitive data upon detecting intrusions.
- The proposed research will employ the Zero Trust Security Model. ZTS model is highly popular in the field of cloud computing and cyber security, in this no user or system is trusted, thus ensuring security of data.
- The proposed system will implement the logging of all the activities and implementation of auditing capabilities; it will ensure the proper management of the system.
- Implementation of automation in our system will ensure the minimization of the human made errors. In this, data will automatically delete as soon as intrusion is detected, thus enhances overall efficiency of the system.
- This research will offer a practical solution to the organisations who deals with highly sensitive and confidential data.

Proposed Solution

- In this research, we proposed a development of an exhaustive system for automated data deletion in a multi-cloud environment with the capabilities of Intrusion detection. The proposed system aims to address the complex challenges associated with managing data privacy and security across multi-cloud infrastructures. The key components of the system include a policy driven data deletion mechanism, an intrusion detection system tailored for multi-cloud architectures, and a centralized policy engine for defining, evaluating, and enforcing policies governing data deletion, access control, and compliance.

Key Concepts

- Cloud Computing
- Data Security
- Intrusion Detection
- Automated Response
- Cyber Security
- Cloud Security
- Research

Outcome

- In this research, the system is proposed which will ensure the security of the data over multi-cloud platforms. It will protect the data from data breaches and mitigate the threats. In this proposed system, data will be automatically deleted as soon as intrusion is detected. This research will increase the confidence in users that their data is well protected in multi-cloud environment. This system is going to encourage the stakeholders to use cloud services instead of traditional systems to store and manage the data. The usage of zero trust model architecture in our system, increases the confidence in the users that their data is well protected and secured. This research will provide a blueprint to the stakeholders to establish, design and implement the data security measures. It will also offer a practical solution to the organisations who deals with highly sensitive and confidential data and maintains high-level security.